

DEVO on EVIMO — Event Filtering: ATE Comparison

Absolute Trajectory Error (cm), sim3 alignment (Umeyama) — EVIMO validation split, 13 sequences

Experimental Setup

Evaluated on the EVIMO validation split (DVS346, 260×346 px, 13 sequences). DEVO network weights are frozen across all runs; only the input event stream varies. ATE is computed with sim3 (Umeyama) alignment to remove monocular scale ambiguity. Four filtering configurations are compared:

- **Original:** Unfiltered events from raw ROS bags. GT poses from Vicon (absolute world frame).
- **Filter v1 (broken):** 3D bounding-box projection with flat rectangle proxies for 3D toy objects. Camera-frame projection had orientation bugs. Net removal: ≈0.3% of events. Same GT as original.
- **Filter NPZ / wrong GT:** Pixel-perfect segmentation masks from EVIMO NPZ files (13.2% events removed). GT derived from NPZ metadata (poses relative to first frame, ~40 Hz). *Not directly comparable to runs 1–2; included to show the GT-bias effect.*
- **Filter NPZ / correct GT:** Same pixel-perfect NPZ masks (13.2% removed), with GT and frame timestamps from the original bags. Fair comparison with run 1.

Results

	Improvement vs. original
	Degradation vs. original
	Best ATE across all four runs

Sequence	Filtered (%)	Original (no filter)	Filter v1 (broken)	Filter NPZ wrong GT*	Filter NPZ correct GT	Δ ATE vs. Original
box / seq_00	1.6%	5.36	3.20	12.34	4.61	−0.75 cm
box / seq_01	5.3%	37.60	38.28	26.23	38.15	+0.55 cm
box / seq_02	8.1%	7.03	6.71	7.60	4.28	−2.75 cm
fast / seq_00	12.1%	11.04	11.63	4.89	10.86	−0.18 cm
fast / seq_01	12.1%	10.48	8.11	13.87	11.31	+0.83 cm
floor / seq_00	10.3%	2.59	2.58	8.79	2.42	−0.17 cm
floor / seq_01	11.7%	2.68	2.55	5.03	2.38	−0.30 cm
table / seq_00	8.8%	4.29	4.50	8.85	4.71	+0.42 cm
table / seq_01	8.1%	4.04	3.30	4.69	2.82	−1.22 cm
tabletop / seq_00	16.0%	12.21	2.27	6.54	2.42	−9.79 cm
tabletop / seq_01	15.4%	1.50	1.60	4.30	3.05	+1.55 cm
tabletop / seq_02	7.3%	6.90	5.18	9.23	4.85	−2.05 cm
tabletop / seq_03	32.4%	0.18	0.18	0.23	0.18	±0.00 cm
Mean (13 seq.)	11.4%	8.14	6.93	7.88	6.32	−1.82 cm
Improvement		—	−14.9%	−3.2%	−22.4%	

* Filter NPZ / wrong GT uses NPZ-relative poses instead of Vicon absolute frame — biased comparison, not used for final conclusions.

Filtering stats (NPZ run): 566,921,089 events total — 75,012,266 removed (13.2%).

Key Findings

- Filter v1 removed only 0.3% of events due to projection bugs, yet mean ATE improved by 14.9% — suggesting DEVO is sensitive to even a small fraction of object-generated events.

- Pixel-perfect NPZ masks remove 13.2% of events (median 10.3% per sequence). With the same GT as the original, this yields a **22.4% reduction in mean ATE (8.14 → 6.32 cm)**.

- 9/13 sequences improve, 3 slightly degrade. The largest gain is tabletop/seq_00 (−9.79 cm, 16% filtered), where the moving object fills a significant portion of the field of view.

- The apparent degradation in Filter NPZ / wrong GT is entirely due to the GT reference mismatch (NPZ-relative vs. Vicon absolute), not to the filtering itself.